

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

The Importance of Data Warehouse Implementation in Any Business



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Understand information about data and its structure in simple spoken and written situations.

Ask your partner:

- According to you, what is the best way to create, store, and manage data information in a company?
- What do you understand by 'raw data'? Share your opinion.

Can Do Statements:

- I can talk about the implementation of a data warehouse.
- I can explain different steps related to the implementation of a data warehouse.
- I can exchange information about data warehouse implementation at a specific context.



Let's Explore!

Let's follow the procedure to achieve the data warehouse implementation.



Instructions: Read the text below.

Jane: John, I need your help. I was distracted during the data warehouse class and I didn't understand anything about the lesson.

John: Don't worry; I worked for a company which managed a tremendous amount of data.

Jane: Great. Can you describe to me the data warehousing implementation?

John: Sure! First of all, data can be used for different purposes such as sales, customer service, or inventory control. Plus, we use these warehouses to save data.

Jane: Right.

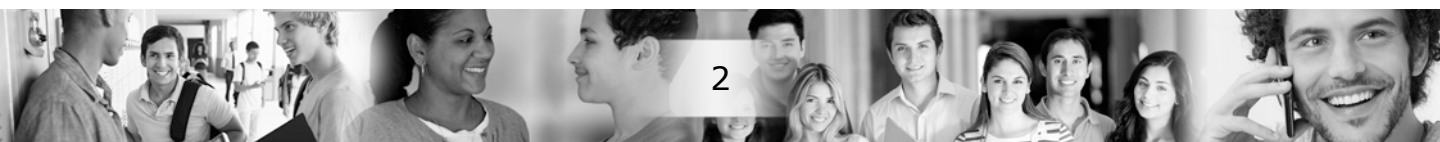
John: Well, the clue when you are designing a data warehouse is that researchers must have a way to use analytical tools to access to the raw data in the database software.

Plus, a data warehouse is closely related to data mining. It can be observed as the exploitation of the collection of views, queries, and other elements that can be generated using the data warehouse as the infrastructure.

Jane: I understood, but can you explain to me the steps of the data warehouse implementation?

John: As I mentioned, the first step is to check the raw data, which is data gathered from any transaction. After that, a "cleaning process" takes place in which a careful validation is done. The third phase is the storage, where decisions as quality, query language, and sources are made. Next, the implementation needs a codification of the structure; here the implementation defines the relationships between query and data. Finally, an analytical tool is set so that the user can have an interaction and support the data during the uploading process.

For example, we can use a Datalog to reexamine transaction data to determine the best way to sell different types of products. In order to help that research, we have to store the collected data and allow it to be queried. All in all, the main purpose of data warehouse is to provide an organized repository for data.





Let's Practice!

Jane is looking for* some vocabulary she needs to talk about the implementation of a data warehouse.

*To "look for" has a similar meaning to that of "search" or "seek".

Instructions: Find the words in the word search below.

1. repository
2. exploitation
3. datalog
4. infrastructure
5. querylanguage

I	D	E	N	O	I	T	A	T	I	O	L	P	X	E	U
N	Q	N	B	O	L	S	I	T	A	U	U	H	G	H	M
F	O	F	N	O	I	T	A	T	C	E	P	X	E	S	E
R	C	R	S	P	G	C	I	L	D	T	B	A	D	C	U
A	Z	E	F	S	V	T	V	C	B	S	I	O	A	L	I
S	M	P	U	Z	M	W	J	G	A	K	D	T	T	C	O
T	K	O	M	R	Y	G	R	E	K	L	G	T	A	K	T
R	E	S	E	V	A	T	E	S	O	V	L	Z	L	X	I
U	M	I	L	D	L	F	L	E	K	B	Q	Y	O	K	L
C	U	T	D	C	O	B	D	M	V	Q	O	Z	G	B	L
T	X	O	A	I	C	I	T	O	S	I	N	G	N	R	I
U	X	R	T	T	E	O	V	T	R	H	E	K	I	R	C
R	O	Y	U	F	L	O	R	E	C	I	T	A	S	P	O
E	E	P	H	N	X	L	M	R	D	Y	I	M	E	V	P
A	E	L	Y	U	Q	U	I	T	A	V	L	Z	D	X	E
B	M	E	G	A	U	G	N	A	L	Y	R	E	U	Q	H



Let's Practice!

John tells Jane about some of the benefits of data warehouse implementation.

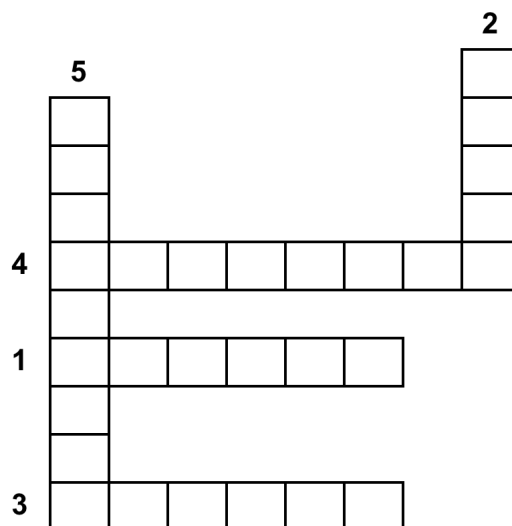
Instructions: Read the clues. Complete the crossword.

Across

1. It is important to d _ _ _ _ a data warehouse for big companies.
3. It is important to d _ _ _ _ a data warehouse for big companies.
4. You can o _ _ _ _ _ data in a data warehouse.

Down

2. You need to be able to s _ _ _ _ data in an organized place.
5. You can have an interaction with data when you use an analytical tool during the u _ _ _ _ _ _ _ _ process.





Extension Activity

Let's find out more about data warehouse!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about implementing data warehouses.
- Make a list of all the additional benefits of data warehouses that were not covered on this learning object.
- Share and discuss your list with a partner.

Self-Assessment

Yes No

- I search for more information regarding data warehouses.
- I list all the benefits of data warehouse implementation.
- I discuss and share my list with a partner.

☐	☐
☐	☐
☐	☐



Target Words

- **analytical tool** (noun): the use of logical reasoning instruments to evaluate data
e.g. Finally, an analytical tool is set so that the user can have an interaction and support the data during the uploading process.
- **Datalog** (noun): programming purely declarative language resulting from Prolog used in deductive database design
e.g. Of course, transaction data might be reexamined with a Datalog to determine the best way to sell different type of products.
- **repository** (noun): place or space where data is stored
e.g. All in all, the main purpose of data warehouse is to provide an organized repository for data.
- **query languages** (noun): languages used to make queries in databases e.g. The third phase is the storage, where decisions as quality, query language, and sources are made.
- **infrastructure** (noun): networks, systems, and processes which contribute to the function of a system e.g. It can be observed as the exploitation of the collection of views, queries, and other elements that can be generated using the data warehouse as the infrastructure.
- **exploitation** (noun): utilization of sources
e.g. It can be observed as the exploitation of the collection of views, queries, and other elements that can be generated using the data warehouse as the infrastructure.



Learning Points

Action Verbs Used in Data Warehouse

There are some **verbs** that you can use in order to express actions associated to the data warehousing processes.

Action verbs are words that explicitly express what the subject is performing. These verbs bring a great deal of information in any sentence.

Different from nouns, adjectives or adverbs which cannot make a one-word sentence.

Let's take a look at the following **action verbs** related to data warehousing.

Verb	Meaning
Design	To prepare a plan to execute any action.
Store	To keep important information in different systems.
Save	To copy pieces of information onto floppy device or hard memory.
Gather	To collect or accumulate pieces of information.
Organize	To form groups of data.
Define	To make an outline of something.
Describe	To draw a material or non-material object, thing or idea.

Examples

- *Well, the clue when you are designing a data warehouse is that researchers must have a way to use analytical tools to access to the raw data in the database software.*
- *In order to help that research, we have to store the collected data allow it to be queried.*





Learning Points

Content Words Used to Talk About Data Warehouses

Content words provide the readers the contents of the story and tell where to focus their attention.

Content words are nouns, verbs, adjectives, and sometimes adverbs that are relevant to a specific topic.

Those are words that give us an image in our minds. In fact, they have the ability to make the reader be able to quickly understand the main idea of a text or topic.

Now, there are some important content words that we have to review to understand more about database warehouses. Let's check some of them:

Category	Description
Query languages	Query languages are basically computer commands that help systems create queries in databases.
Analytical tools	A series of tools that process data and information.
Raw data	Also known as 'primary data'. It is basically the unprocessed data that is obtained from a specific source.
Collection	A series of elements.
Infrastructure	The framework of a system like a database.

Examples

- *The third phase is the storage, where decisions as quality, query language, and sources are made.*
- *Well, the clue when you are designing a data warehouse is that researchers must have a way to use analytical tools to access to the raw data in the database software.*



Learning Points

List of Programming and Computer Science Terms

These are **words you can hear when you work in a programming company**, they are computer science global terms.

Many of the Computer science words are related to programming; you can use them to express the terms appropriately.

Word	Meaning
Daemon	Processes that works in the background of any operating system.
Datalog	Programming purely declarative language resulting from Prolog used in deductive database design.
DCOM	(Distributed Component Object Model) It provides interfaces which allow clients and servers to communicate with the computer.
DDT	Dynamic Debugging Technique
Dead Code	Unnecessary section of the source code.